MD.ABID ULLAH MUHIB

Lecturer, Computer Science and Engineering

Sylhet, Bangladesh (UTC+6) | uusshas12@gmail.com | +880 1859-172800 | imusshas.github.io

linkedin.com/in/md-abid-ullah-muhib-702862417 | github.com/imusshas

scholar.google.com/citations?user=tXLhGl4AAAAJ

SUMMARY

Computer Science and Engineering (CSE) Lecturer candidate and researcher with a BSc from SUST (CGPA 3.80/4.00, Rank 7th/110). Authored five peer-reviewed IEEE publications and one journal manuscript at PLOS ONE. Built end-to-end PyTorch deep learning pipelines for Transformer and protein language models using LoRA, DPO, and KTO. Proficient in C, C++, Python, and Java with 700+ solved algorithmic problems and certified as an AWS Solutions Architect (Associate).

SKILLS

- **Machine Learning & Deep Learning**: PyTorch, Hugging Face Transformers, PEFT, LoRA, SFT, DPO, KTO, Contrastive Learning, Model Training, Fine-Tuning, Model Inference, Evaluation, Benchmarking, Error Analysis, scikit-learn
- **Core Computing & Instruction**: Data Structures and Algorithms, Structured Programming (C), Object-Oriented Programming (C++/Java), Database Management Systems, Operating Systems, Computer Networks, Computer Architecture
- **Programming Languages**: C, C++, Python, Java, SQL, TypeScript, Bash
- **Systems, Cloud & Infrastructure**: AWS Certified Solutions Architect (Associate), Docker, Linux, Git, GitHub, Spring Cloud, Next.js, PostgreSQL, MySQL, Redis, Lima Virtualization, Qdrant Vector DB, CI/CD
- **Domains & Specializations**: Natural Language Processing (NLP), Protein Language Models, Computational Biology, Computer Vision, Low-Resource Languages, Dataset Curation, Seismic Signal Modeling, Remote Sensing

WORK EXPERIENCE

Research Assistant, Department of Computer Science and Engineering, SUST 2025 – Present

Sylhet, Bangladesh

- Investigated geographic training distribution shifts across the 65-scene Sentinel-1 (S1S2 Water) satellite benchmark.
- Proposed a scene-uniform exposure intervention, improving site-macro IoU from 0.500 to 0.541 across 73 global benchmark sites over ERM.
- Documented the research and compared error analyses across 5 baseline architectures.

EDUCATION

Bachelor of Science (BSc), Computer Science and Engineering Mar 2022 – Jul 2026

Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh

- **Academic Standing**: CGPA 3.80 / 4.00 (Class Rank: 7th out of 110 students; Top 6.5% of graduating cohort).
- **Undergraduate Capstone Research**: Parameter-efficient fine-tuning (LoRA) and preference optimization (DPO/KTO) of ProGen2 for thermophile sequences, achieving a 76.56% win rate (under review at PLOS ONE).
- **Coursework**: Data Structures, Algorithms, Structured Programming, OOP, Database Systems, Computer Networks, Operating Systems, Computer Architecture, Machine Learning, Cloud Computing.

PUBLICATIONS

Parameter-Efficient Preference Optimization and Supervised Fine-Tuning of ProGen2 for Thermophile-Associated Protein Sequence Space Aug 2025 – Present

PLOS ONE (Under Review)

- Tailored a homology-controlled benchmark from Learn2ThermDB containing 224,646 thermophile sequences and 49,399 pairs.
- Fine-tuned the 151M-parameter ProGen2 model with rank-8/16 LoRA, achieving a 76.56% win rate via Direct Preference Optimization (DPO).
- Demonstrated that Supervised Fine-Tuning (SFT) minimized absolute amino-acid-conditioned negative log-likelihood to 2.6019.

Real-Time Earthquake Magnitude Classification from Initial P-Waves: Models, Dataset, and Comparative Analysis for South Asia IEEE ICCIT 2025

doi:10.1109/ICCIT68739.2025.11489542

- Curated a standardized benchmark of 7,318 South Asian seismic events using 7-second P-wave windows for early warning.
- Built a Deep Attention Transformer achieving 81.56% adaptive accuracy with 4.8 ms real-time inference latency.
- Introduced an Adaptive Accuracy metric (± 0.2 tolerance), proving that 60.3% of model errors concentrate at class boundaries.

Protein-Based Fish Species Identification: Dataset, Models, and Insights from Native Bangladeshi Fish IEEE QPAIN 2026
doi:10.1109/QPAIN69676.2026.11546620
- Introduced the first benchmark of 2,845 high-quality protein sequences across 9 native Bangladeshi fish species.
- Designed a MotifCNN-Transformer architecture with Terminal-Aware Positional Encoding, achieving 79.80% accuracy.
- Optimized model efficiency to be 42× smaller and 5× faster than ProtBERT, enabling real-time GPU-free inference.

A Novel Transformer-Based Framework for Bangla Tense Classification IEEE QPAIN 2026
doi:10.1109/QPAIN69676.2026.11546186
- Benchmarked 5 transformer architectures on the 17,819-sentence annotated BanglaTense benchmark dataset.
- Established a new state of the art with BGE-M3 embeddings, achieving 99.49% accuracy and 99.50% F1-score.
- Improved classification performance over previous Recurrent Neural Network (RNN) baselines by 3.57% absolute accuracy.

BanglaShadhu: A Contrastive Transformer-Based Approach for Distinguishing Sadhu Bhasha and Cholito Bhasha in Bengali Text IEEE QPAIN 2026
doi:10.1109/QPAIN69676.2026.11546481
- Evaluated transformer models on the balanced 7,350-sentence BanglaBlend dataset to separate formal and colloquial registers.
- Achieved 96.19% accuracy using BGE-M3 contrastive embeddings, significantly outperforming language-specific BERT models.
- Analyzed cross-lingual semantic transfer across 4 model variants, verifying robust separation across diverse syntactic structures.

Transformer-Based Hate Speech Detection in Bangla Regional Dialects: A Novel Framework for Chittagong, Barishal, and Noakhali Dialects IEEE QPAIN 2026
doi:10.1109/QPAIN69676.2026.11545860
- Examined transformer classifiers on the BIDWESH dataset of 9,179 dialectal samples across 3 low-resource regional dialects.
- Reached a state of the art 95.97% accuracy and F1-score using contrastive BGE-M3 embeddings.
- Outperformed traditional masked language modeling baselines by 4.2% across Chittagong, Barishal, and Noakhali dialects.

PROJECTS

Infrastructure-as-a-Service (IaaS) Platform May 2026 – Jun 2026
- Architected an on-premises compute orchestration engine in Next.js 16 and TypeScript, automating virtualization provisioning across 4 Linux distributions via Lima (`limactl`) with sub-10-second initialization.
- Implemented dynamic hardware capacity telemetry via `systeminformation`, reserving 2048 MB host RAM and enforcing 20 GB storage quotas across multi-core virtual machines.
- Developed a real-time Server-Sent Events (SSE) control plane delivering chunked download progress at 5% intervals and live lifecycle telemetry to optimistic React 19 UI components.

Nikhoj: Multi-Modal Lost-and-Found Platform Jul 2025 – Sep 2025
- Planned the distributed system architecture connecting 6+ Spring Cloud microservices and a PostgreSQL database to Qdrant Vector DB for multi-modal similarity search.
- Integrated dual deep learning pipelines extracting 512-dimensional facial embeddings via FaceNet-512 and multilingual semantic text representations via BGE-M3.
- Engineered the responsive React and TanStack client, building interactive query workflows that achieve sub-50 ms candidate similarity ranking across lost-and-found records.

ACHIEVEMENTS

- Earned the AWS Certified Solutions Architect – Associate certification and AWS Academy Graduate – Cloud Developing training badge.
- Solved 700+ algorithmic problems across competitive programming platforms; Preliminary Round Champion at Hack The AI 2025 and Finalist at CUET ETEverse 2025.
- Served as Technical Executive of the SUST ACM Student Chapter and Executive Member of the CSE Society, SUST (2025-26).
- Semi-Finalist in the Inter-Cantonment Parliamentary Debate Competition (Dhaka Cantonment Girls' Public School and College, 2018), Runner-up in Annual Debate Competition (Cantonment College Jashore, 2019), and Competitor at the Rotaract District Science Fair (JUST, 2019).

REFERENCES

- **Dr. Mohammad Shahidur Rahman**, Professor, Department of CSE, SUST.
  Email: rahmanms@sust.edu
- **Dr. Md Forhad Rabbi**, Professor, Department of CSE, SUST.
  Email: frabbi-cse@sust.edu